

4.	D	The force on the racket must be equal and opposite to the force on the ball based on Newton's 3rd Law.
5.	A	Acceleration along slope: $a = g \sin 10^\circ = 1.7$ velocity of block after 3.0 s: $v = u + at = 0 + 1.7(3) = 5.1$ momentum $p = mv = 1.8 \times 5.1 = 9.18 \text{ kg m s}^{-1}$
6.	B	At maximum height initial horizontal velocity $= 50 \cos 80^\circ = 8.62 \text{ m s}^{-1}$